

Lecture 3: Geometry of OLS and partitioned regression

Economics 527: Econometric Methods

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Geometry of OLS

Least squares as a distance

- OLS, first lecture:

$$\hat{\beta} = \arg \min_{b \in \mathbb{R}^k} \underbrace{(Y - Xb)^\top (Y - Xb)}_{= \|Y - Xb\|^2}$$

- Norm of $a \in \mathbb{R}^n$:

$$\begin{aligned}\|a\| &= \sqrt{\sum_{i=1}^n a_i^2} \\ &= \sqrt{a^\top a}\end{aligned}$$

Column space $S(X)$

- Column space of X :

$$S(X) = \left\{ v \in \mathbb{R}^n : v = \underbrace{X}_{n \times k} \underbrace{b}_{k \times 1} \text{ for some } b \in \mathbb{R}^k \right\}$$

- Partition X column-wise:

$$\underbrace{X}_{n \times k} = \left[\underbrace{\underline{X}_1}_{n \times 1} \cdots \underbrace{\underline{X}_k}_{n \times 1} \right],$$

where $\underline{X}_j$ is the $n \times 1$ vector of the observations on the j -th regressor.

- What the elements of $S(X)$ look like, in terms of the columns of X :

$$\begin{aligned}\underbrace{\underline{X}}_{n \times k} \underbrace{b}_{k \times 1} &= \underbrace{[\underline{X}_1 \cdots \underline{X}_k]}_{=X} \begin{bmatrix} b_1 \\ \vdots \\ b_k \end{bmatrix} \\ &= b_1 \underline{X}_1 + \cdots + b_k \underline{X}_k \\ &= \sum_{j=1}^k b_j \underline{X}_j\end{aligned}$$

- Each $\underline{X}_j \in S(X)$, since we can take $b_j = 1$, $b_i = 0$ for $i \neq j$:

$$\begin{aligned}Xb &= [\underline{X}_1 \cdots \underline{X}_k] \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \\ &= \underline{X}_j \in S(X)\end{aligned}$$

- $S(X) \subset \mathbb{R}^n$: k -dimensional subspace; $\text{rank}(X) = k < n$.

$\hat{Y}$: closest to Y in $S(X)$

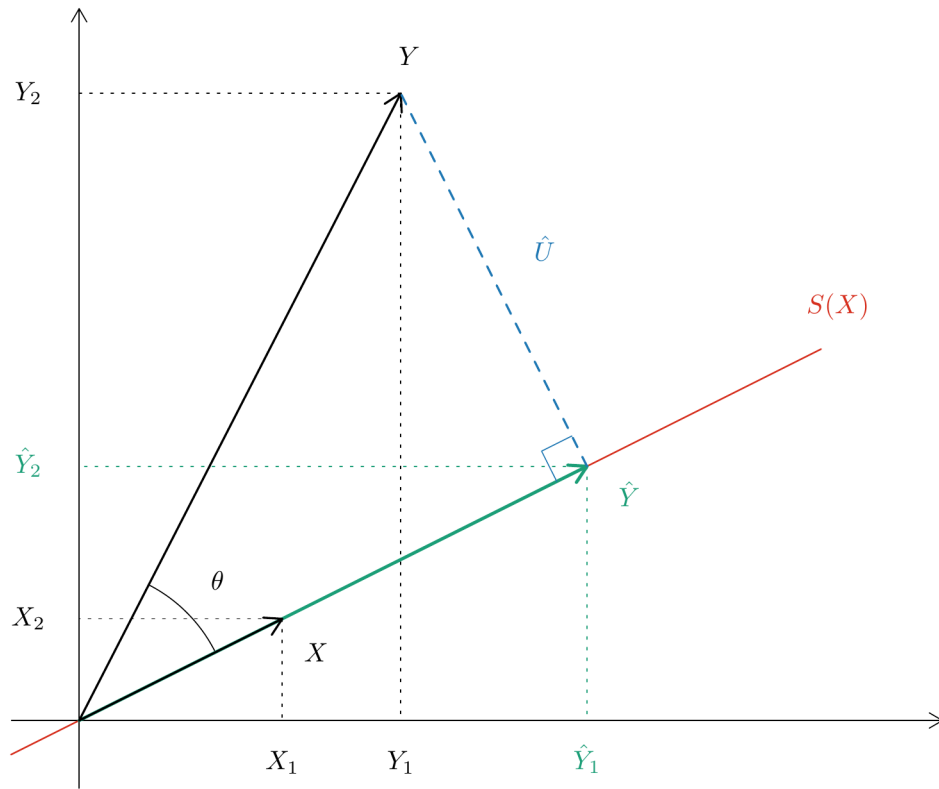
- We can re-state the OLS problem in terms of $S(X)$:

$$\hat{\beta} = \arg \min_{b \in \mathbb{R}^k} \|Y - Xb\|^2, \quad \hat{Y} = X\hat{\beta}$$

$\Leftrightarrow$

$$\hat{Y} = \arg \min_{v \in S(X)} \|Y - v\|^2$$

- When $k = 1$, $n = 2$, we can visualize the solution: You can find the closest element by dropping a perpendicular from Y to $S(X)$.
- $Y = (Y_1, Y_2)^\top$, $X = (X_1, X_2)^\top$, and $S(X) = \{(bX_1, bX_2)^\top : b \in \mathbb{R}\}$.



- The coefficient $\hat{\beta}$ is determined by the angle θ between X and Y :

$$\begin{aligned} \cos \theta &= \frac{X^\top Y}{\|X\| \|Y\|} \\ \hat{\beta} &= \frac{X^\top Y}{\|X\|^2} \\ &= \cos \theta \cdot \frac{\|Y\|}{\|X\|} \end{aligned}$$

- Residuals, perpendicular to $S(X)$:

$$\hat{U} = Y - \hat{Y} = Y - X\hat{\beta}$$

Dropping the perpendicular: the algebra

- Fitted values:

$$\begin{aligned}\hat{Y} &= X\hat{\beta} \\ &= \underbrace{X(X^\top X)^{-1}X^\top}_{P_X} Y \\ &= P_X Y\end{aligned}$$

- Residuals:

$$\begin{aligned}\hat{U} &= Y - \hat{Y} \\ &= Y - P_X Y \\ &= \underbrace{(I_n - P_X)}_{= M_X} Y \\ &= M_X Y\end{aligned}$$

- Both $n \times n$:

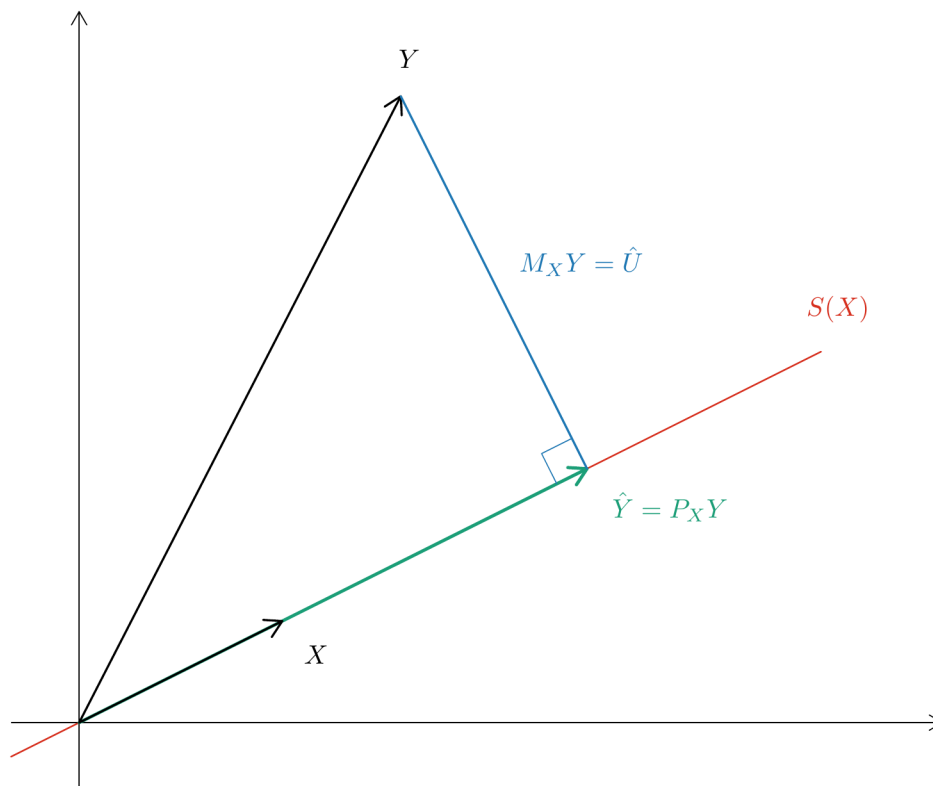
$$\boxed{P_X = \underbrace{X}_{n \times k} \underbrace{(X^\top X)^{-1}}_{k \times k} \underbrace{X^\top}_{k \times n} \quad M_X = \underbrace{I_n}_{n \times n} - \underbrace{P_X}_{n \times n}}$$

Projection matrices

$P_X Y$ and $M_X Y$

- Decomposition of $Y \in \mathbb{R}^n$:

$$\begin{aligned}P_X + M_X &= I_n \\ \Rightarrow P_X Y + M_X Y &= Y\end{aligned}$$



Properties of P_X

- Symmetric, by $(AB)^\top = B^\top A^\top$:

$$\begin{aligned} P_X^\top &= (X(X^\top X)^{-1}X^\top)^\top \\ &= X(X^\top X)^{-1}X^\top \\ &= P_X \end{aligned}$$

- $P_X X = X$:

$$\begin{aligned} P_X X &= \underbrace{X(X^\top X)^{-1}X^\top}_{P_X} X \\ &= X \end{aligned}$$

- Column by column, $P_X \underline{X}_j = \underline{X}_j$ for $j = 1, \dots, k$:

$$\begin{aligned} P_X X &= P_X \begin{bmatrix} \underline{X}_1 & \cdots & \underline{X}_k \end{bmatrix} \\ &= \begin{bmatrix} P_X \underline{X}_1 & \cdots & P_X \underline{X}_k \end{bmatrix} \\ &= \begin{bmatrix} \underline{X}_1 & \cdots & \underline{X}_k \end{bmatrix}, \end{aligned}$$

where the last equality holds because $P_X X = X$.

- Idempotent:

$$\begin{aligned} P_X^2 &= P_X P_X \\ &= \underbrace{P_X X}_{=X} (X^\top X)^{-1} X^\top \\ &= P_X \end{aligned}$$

P_X : p.s.d., trace k

- For $a \in \mathbb{R}^n$:

$$\begin{aligned} a^\top P_X a &= \underbrace{a^\top X}_{w^\top} \underbrace{(X^\top X)^{-1}}_{\text{p.d.}} \underbrace{X^\top a}_w \\ &\geq 0 \end{aligned}$$

- We will use: $\text{tr}(AB) = \text{tr}(BA)$.

$$\begin{aligned} \text{tr}(P_X) &= \text{tr}(X(X^\top X)^{-1}X^\top) \\ &= \text{tr}\left((X^\top X)^{-1} \underbrace{X^\top X}_{\substack{k \times n \quad n \times k \\ k \times k}}\right) \\ &= \text{tr}(I_k) \\ &= k \end{aligned}$$

- P_X : $n \times n$, symmetric, p.s.d., idempotent, rank k .
- $P_X : \mathbb{R}^n \rightarrow S(X)$.

Properties of M_X

- Symmetric:

$$\begin{aligned} M_X^\top &= I_n^\top - P_X^\top \\ &= I_n - P_X \\ &= M_X \end{aligned}$$

- $M_X X = 0$:

$$\begin{aligned} M_X X &= (I_n - P_X)X \\ &= X - \underbrace{P_X X}_{=X} \\ &= X - X \\ &= 0 \end{aligned}$$

- $v = Xb \in S(X)$, $b \in \mathbb{R}^k$:

$$\begin{aligned} M_X v &= M_X Xb \\ &= 0 \end{aligned}$$

$M_X P_X = 0$ and idempotency

- $M_X P_X = 0$ and $P_X M_X = 0$:

$$\begin{aligned} M_X P_X &= \underbrace{M_X X}_{=0} (X^\top X)^{-1} X^\top \\ &= 0 \\ P_X M_X &= (M_X P_X)^\top \\ &= 0 \end{aligned}$$

- Idempotent:

$$\begin{aligned} M_X M_X &= (I_n - P_X)M_X \\ &= M_X - \underbrace{P_X M_X}_{=0} \\ &= M_X \end{aligned}$$

Orthogonal complement $S^\perp(X)$

- Orthogonal complement of $S(X)$:

$$S^\perp(X) = \{v \in \mathbb{R}^n : v^\top X = 0\}$$

- Normal equations, first lecture:

$$\begin{aligned} X^\top \hat{U} &= X^\top (Y - X\hat{\beta}) \\ &= X^\top Y - X^\top X \underbrace{\hat{\beta}}_{=(X^\top X)^{-1} X^\top Y} \\ &= X^\top Y - X^\top Y \\ &= 0 \\ \implies \hat{U}^\top X &= 0 \\ \implies \hat{U} &= M_X Y \in S^\perp(X) \end{aligned}$$

- $M_X : \mathbb{R}^n \longrightarrow S^\perp(X)$.

M_X : rank $n - k$, p.s.d.

- Rank:

$$\begin{aligned} \text{rank}(M_X) &= \text{tr}(M_X) \\ &= \text{tr}(I_n - P_X) \\ &= \text{tr}(I_n) - \text{tr}(P_X) \\ &= n - k \end{aligned}$$

- M_X : $n \times n$; no inverse.
- **Claim.** M_X : p.s.d. For $a \in \mathbb{R}^n$ and $w = M_X a$:

$$\begin{aligned}
a^\top M_X a &= a^\top M_X M_X a \quad (\text{idempotent}) \\
&= \underbrace{a^\top M_X^\top}_{w^\top} \underbrace{M_X a}_w \quad (\text{symmetric}) \\
&= w^\top w \\
&= \|w\|^2 \\
&\geq 0
\end{aligned}$$

$$Y = \hat{Y} + \hat{U}$$

- Decomposition:

$$\underset{\in \mathbb{R}^n}{Y} = \underset{\in S(X)}{\hat{Y}} + \underset{\in S^\perp(X)}{\hat{U}}, \quad S(X) + S^\perp(X) = \mathbb{R}^n$$

- Fitted values:

$$\begin{aligned}
\hat{Y} &= X\hat{\beta} \\
&= P_X Y \in S(X), \quad \text{dimension } k
\end{aligned}$$

- Residuals:

$$\begin{aligned}
\hat{U} &= Y - \hat{Y} \\
&= M_X Y \in S^\perp(X), \quad \text{dimension } n - k
\end{aligned}$$

- Orthogonality:

$$\begin{aligned}
\hat{Y}^\top \hat{U} &= Y^\top P_X^\top M_X Y \\
&= Y^\top \underbrace{P_X M_X}_{=0} Y \\
&= 0
\end{aligned}$$

Estimation of σ^2

Two estimators of σ^2

- Assumptions 2 and 3:

$$\begin{aligned}
\text{Var}(U \mid X) &= \text{E}[UU^\top \mid X] \\
&= \sigma^2 I_n
\end{aligned}$$

- Unknown parameter:

$$\begin{aligned}
\sigma^2 &= \text{E}[U_i^2 \mid X] \\
&= \text{E}[U_i^2]
\end{aligned}$$

- Estimators:

$$\begin{aligned}
\hat{\sigma}^2 &= \frac{1}{n} \sum_{i=1}^n \hat{U}_i^2 \\
&= \frac{1}{n} \sum_{i=1}^n (Y_i - X_i^\top \hat{\beta})^2 \\
&= \frac{1}{n} \hat{U}^\top \hat{U} \\
s^2 &= \frac{1}{n-k} \sum_{i=1}^n \hat{U}_i^2
\end{aligned}$$

- **Claim.** Under Assumptions 1 to 4:

$$\mathbb{E}[s^2 | X] = \sigma^2$$

Proof, step 1: residuals to errors

- Residuals:

$$\begin{aligned}\hat{U} &= M_X Y \\ &= M_X (X\beta + U) \quad (\text{Assumption 1}) \\ &= M_X U \quad (M_X X = 0)\end{aligned}$$

- Dimensions:

$$\hat{U}_{n \times 1} = M_X \underset{n \times n}{U}_{n \times 1}, \quad \text{rank}(M_X) = n - k$$

- Sum of squared residuals:

$$\begin{aligned}\sum_{i=1}^n \hat{U}_i^2 &= \|\hat{U}\|^2 \\ &= \hat{U}^\top \hat{U} \\ &= (M_X U)^\top (M_X U) \\ &= U^\top M_X^\top M_X U \\ &= U^\top M_X U \quad (\text{symmetric, idempotent})\end{aligned}$$

Proof, step 3: the trace trick

- The quadratic form $U^\top M_X U$ is a scalar:

$$\overbrace{U^\top M_X U}^{1 \times 1}$$

$1 \times n \quad n \times n \quad n \times 1$

- Treat any scalar c as a 1×1 matrix:

$$c = \text{tr}(c).$$

- We have:

$$\begin{aligned}\mathbb{E}[\hat{U}^\top \hat{U} | X] &= \mathbb{E}[U^\top M_X U | X] \\ &= \mathbb{E}[\text{tr}(\textcolor{red}{U}^\top M_X U) | X] \quad (\text{scalar} = \text{its trace}) \\ &= \mathbb{E}[\text{tr}(M_X U \textcolor{red}{U}^\top) | X] \quad (\text{tr}(AB) = \text{tr}(BA)) \\ &= \text{tr}(\mathbb{E}[M_X U U^\top | X]) \\ &= \text{tr}(M_X \underbrace{\mathbb{E}[U U^\top | X]}_{= \sigma^2 I_n}) \\ &= \sigma^2 \text{tr}(M_X) \\ &= \sigma^2(n - k)\end{aligned}$$

$\hat{\sigma}^2$: biased

- Residuals:

$$\begin{aligned}\mathbb{E}\left[\sum_{i=1}^n \hat{U}_i^2 | X\right] &= \sigma^2(n - k) \\ \Rightarrow \mathbb{E}\left[\sum_{i=1}^n \hat{U}_i^2\right] &= \sigma^2(n - k) \quad (\text{LIE})\end{aligned}$$

- Errors:

$$\begin{aligned} \mathbb{E}[U^\top U] &= \mathbb{E}\left[\sum_{i=1}^n U_i^2\right] \\ &= n\sigma^2 \end{aligned}$$

- Biased, by a known factor:

$$\begin{aligned} \mathbb{E}[\hat{\sigma}^2 \mid X] &= \frac{1}{n} \mathbb{E}[\hat{U}^\top \hat{U} \mid X] \\ &= \frac{n-k}{n} \sigma^2 \\ &\neq \sigma^2 \end{aligned}$$

s^2 : unbiased

- Correction:

$$\begin{aligned} \frac{n}{n-k} \mathbb{E}[\hat{\sigma}^2 \mid X] &= \sigma^2 \\ s^2 &= \frac{n}{n-k} \hat{\sigma}^2 \\ &= \frac{\hat{U}^\top \hat{U}}{n-k} \\ &= \frac{\sum_{i=1}^n \hat{U}_i^2}{n-k} \\ &= \frac{\sum_{i=1}^n (Y_i - X_i^\top \hat{\beta})^2}{n-k} \\ &= \frac{\|Y - X\hat{\beta}\|^2}{n-k} \end{aligned}$$

- Unbiased:

$$\boxed{\mathbb{E}[s^2 \mid X] = \sigma^2}$$

$$\stackrel{\text{LIE}}{\implies} \mathbb{E}[s^2] = \sigma^2$$

Standard errors

- Second lecture: $\text{Var}(\hat{\beta} \mid X) = \sigma^2 (X^\top X)^{-1}$. Estimated:

$$\widehat{\text{Var}}(\hat{\beta} \mid X) = s^2 (X^\top X)^{-1}$$

- Standard error against standard deviation:

$$\begin{aligned} \text{std. err}(\hat{\beta}_j) &= \sqrt{s^2 [(X^\top X)^{-1}]_{jj}} \\ &= \sqrt{\widehat{\text{Var}}(\hat{\beta}_j \mid X)} \\ \text{std. dev}(\hat{\beta}_j) &= \sqrt{\text{Var}(\hat{\beta}_j \mid X)} \end{aligned}$$

Partitioned regression

Partition

- Two blocks of regressors, $k_1 + k_2 = k$:

$$X_{n \times k} = \begin{bmatrix} X_1 & X_2 \\ n \times k_1 & n \times k_2 \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} \begin{matrix} k_1 \times 1 \\ k_2 \times 1 \end{matrix}$$

- The model can be written as

$$\begin{aligned} Y &= \begin{bmatrix} X_1 & X_2 \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} + U \\ &= X_1\beta_1 + X_2\beta_2 + U \end{aligned}$$

- OLS:

$$\begin{aligned} \hat{\beta}_{k \times 1} &= \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} \\ &= (X^\top X)^{-1} X^\top Y \end{aligned}$$

Frisch-Waugh-Lovell (FWL) theorem

- **Claim.**

$$\hat{\beta}_1 = (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 Y, \quad M_2 = I_n - P_2, \quad P_2 = X_2 (X_2^\top X_2)^{-1} X_2^\top$$

- M_2 : symmetric, idempotent, $M_2 X_2 = 0$.
- Similarly:

$$\hat{\beta}_2 = (X_2^\top M_1 X_2)^{-1} X_2^\top M_1 Y, \quad M_1 = I_n - X_1 (X_1^\top X_1)^{-1} X_1^\top.$$

Proof

- We will use:

$$\begin{aligned} \hat{U} &= Y - X\hat{\beta} \\ &= Y - X_1\hat{\beta}_1 - X_2\hat{\beta}_2 \\ \Leftrightarrow Y &= X\hat{\beta} + \hat{U} \\ &= X_1\hat{\beta}_1 + X_2\hat{\beta}_2 + \hat{U} \end{aligned}$$

- We have:

$$\begin{aligned} &(X_1^\top M_2 X_1)^{-1} X_1^\top M_2 Y \\ &= (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 (X_1\hat{\beta}_1 + X_2\hat{\beta}_2 + \hat{U}) \\ &= \underbrace{(X_1^\top M_2 X_1)^{-1} X_1^\top M_2 X_1}_{=I_{k_1}} \hat{\beta}_1 \\ &\quad + (X_1^\top M_2 X_1)^{-1} X_1^\top \underbrace{M_2 X_2}_{=0} \hat{\beta}_2 \\ &\quad + (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 \hat{U} \\ &= \hat{\beta}_1 + (X_1^\top M_2 X_1)^{-1} X_1^\top \underbrace{M_2 \hat{U}}_{=\hat{U}} \\ &= \hat{\beta}_1 + (X_1^\top M_2 X_1)^{-1} \underbrace{X_1^\top \hat{U}}_{=0} \\ &= \boxed{\hat{\beta}_1} \end{aligned}$$

$$M_2 \hat{U} = \hat{U}$$

- Normal equations, block by block:

$$\begin{aligned} 0 &= X^\top \hat{U} \\ &= \begin{bmatrix} X_1 & X_2 \end{bmatrix}^\top \hat{U} \\ &= \begin{bmatrix} X_1^\top \\ X_2^\top \end{bmatrix} \hat{U} \\ &= \begin{bmatrix} X_1^\top \hat{U} \\ X_2^\top \hat{U} \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned}$$

- Hence:

$$\begin{aligned} M_2 \hat{U} &= (I_n - P_2) \hat{U} \\ &= \hat{U} - P_2 \hat{U} \\ &= \hat{U} - X_2 (X_2^\top X_2)^{-1} \underbrace{X_2^\top \hat{U}}_{=0} \\ &= \hat{U} \end{aligned}$$

- Hence:

$$\begin{aligned} X_1^\top M_2 \hat{U} &= X_1^\top \hat{U} \\ &= 0 \end{aligned}$$

Residuals on residuals

- We will use: $M_2^\top = M_2$ and $M_2 M_2 = M_2$.
- We have:

$$\begin{aligned} \hat{\beta}_1 &= (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 Y \\ &= (X_1^\top M_2^\top M_2 X_1)^{-1} X_1^\top M_2^\top M_2 Y \\ &= ((M_2 X_1)^\top (M_2 X_1))^{-1} (M_2 X_1)^\top (M_2 Y) \\ &= (\tilde{X}_1^\top \tilde{X}_1)^{-1} \tilde{X}_1^\top \tilde{Y} \end{aligned}$$

- $\tilde{X}_1 = M_2 X_1$, $\tilde{Y} = M_2 Y$, $\tilde{U} = M_2 U$:

$$\begin{aligned} Y &= X_1 \beta_1 + X_2 \beta_2 + U \\ M_2 Y &= M_2 X_1 \beta_1 + \underbrace{M_2 X_2}_{=0} \beta_2 + M_2 U \\ &= M_2 X_1 \beta_1 + M_2 U \\ \tilde{Y} &= \tilde{X}_1 \beta_1 + \tilde{U} \end{aligned}$$

$\tilde{X}_1$ and $\tilde{Y}$: OLS residuals

- Residuals from the regression of the columns of X_1 against X_2 :

$$\begin{aligned}
 \tilde{X}_1 &= M_2 X_1 \\
 &= (I_n - P_2) X_1 \\
 &= X_1 - P_2 X_1 \\
 &= X_1 - X_2 \underbrace{(X_2^\top X_2)^{-1} X_2^\top X_1}_{\hat{\gamma}} \\
 &= X_1 - X_2 \hat{\gamma} \\
 \Leftrightarrow X_1 &= X_2 \hat{\gamma} + \tilde{X}_1
 \end{aligned}$$

- $\tilde{X}_1^\top X_2 = 0$.
- Residuals from the regression of Y against X_2 :

$$\begin{aligned}
 \tilde{Y} &= M_2 Y \\
 &= Y - X_2 \underbrace{(X_2^\top X_2)^{-1} X_2^\top Y}_{\hat{\eta}} \\
 \Leftrightarrow Y &= X_2 \hat{\eta} + \tilde{Y}
 \end{aligned}$$

Regression in three steps

- Remove from X_1 the part collinear with X_2 : $\tilde{X}_1$.
 - Remove from Y the part collinear with X_2 : $\tilde{Y}$.
 - $\hat{\beta}_1$: the OLS coefficient from the regression of $\tilde{Y}$ against $\tilde{X}_1$.
- $\hat{\beta}_1$: the estimated effect of X_1 after controlling for X_2 .
 - Step 2 optional:

$$\begin{aligned}
 \hat{\beta}_1 &= (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 Y \\
 &= (X_1^\top M_2^\top M_2 X_1)^{-1} X_1^\top M_2^\top Y \\
 &= ((M_2 X_1)^\top (M_2 X_1))^{-1} (M_2 X_1)^\top Y \\
 &= (\tilde{X}_1^\top \tilde{X}_1)^{-1} \tilde{X}_1^\top Y
 \end{aligned}$$

- Step 1 is not optional:

$$(X_1^\top X_1)^{-1} X_1^\top \tilde{Y} = (X_1^\top X_1)^{-1} X_1^\top M_2 Y \neq (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 Y \quad \text{in general}$$

Partitioned regression and the intercept

Intercept as a regressor

- One regressor and an intercept:

$$Y_i = \underset{1 \times 1}{\beta_1} + \underset{1 \times 1}{\beta_2} \underset{1 \times 1}{X_{i,2}} + U_i$$

- Matrix form, with ℓ the $n \times 1$ vector of ones:

$$\begin{aligned}
 X &= \begin{bmatrix} 1 & X_{1,2} \\ \vdots & \vdots \\ 1 & X_{n,2} \end{bmatrix} \\
 &= [\ell \quad X_2] \\
 Y &= \underset{1 \times 1}{\beta_1} \underset{n \times 1}{\ell} + \underset{1 \times 1}{\beta_2} \underset{n \times 1}{X_2} + U
 \end{aligned}$$

- Partitioned regression:

$$\begin{aligned}\hat{\beta}_2 &= \underset{1 \times 1}{(X_2^\top M_\ell X_2)^{-1}} \underset{1 \times n}{X_2^\top} \underset{n \times 1}{M_\ell Y} \\ M_\ell &= I_n - \ell(\ell^\top \ell)^{-1} \ell^\top\end{aligned}$$

P_ℓ and M_ℓ

- $\ell^\top \ell$:

$$\begin{aligned}\ell^\top \ell &= \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}^\top \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \\ &= \sum_{i=1}^n 1 \cdot 1 \\ &= n\end{aligned}$$

- Hence:

$$\begin{aligned}P_\ell &= \ell(\ell^\top \ell)^{-1} \ell^\top \\ &= \frac{\ell \ell^\top}{n} \\ M_\ell &= I_n - \frac{\ell \ell^\top}{n}\end{aligned}$$

- For $v = (v_1, \dots, v_n)^\top$:

$$\begin{aligned}(\ell^\top \ell)^{-1} \ell^\top v &= \frac{\ell^\top v}{n} \\ &= \frac{\sum_{i=1}^n \ell_i v_i}{n} \\ &= \frac{\sum_{i=1}^n v_i}{n} \\ &= \bar{v}\end{aligned}$$

P_ℓ : averaging

- For $v \in \mathbb{R}^n$:

$$\begin{aligned}P_\ell v &= \ell(\ell^\top \ell)^{-1} \ell^\top v \\ &= \frac{\ell \ell^\top}{n} v \\ &= \ell \left(\frac{\sum_{i=1}^n 1 \cdot v_i}{n} \right) \\ &= \ell \bar{v} \\ &= \begin{pmatrix} \bar{v} \\ \vdots \\ \bar{v} \end{pmatrix}\end{aligned}$$

- Column space of ℓ :

$$\begin{aligned}S(\ell) &= \{ w \in \mathbb{R}^n : w = c \ell, \ c \in \mathbb{R} \} \\ &= \left\{ w \in \mathbb{R}^n : w = \begin{pmatrix} c \\ \vdots \\ c \end{pmatrix}, \ c \in \mathbb{R} \right\}\end{aligned}$$

M_ℓ : deviations from the mean

- For $v \in \mathbb{R}^n$:

$$\begin{aligned} M_\ell v &= v - P_\ell v \\ &= v - \ell \bar{v} \\ &= \begin{pmatrix} v_1 - \bar{v} \\ \vdots \\ v_n - \bar{v} \end{pmatrix} \end{aligned}$$

- Orthogonal complement of $S(\ell)$:

$$\begin{aligned} S^\perp(\ell) &= \left\{ v \in \mathbb{R}^n : \underbrace{v^\top \ell}_{= \sum_{i=1}^n v_i} = 0 \right\} \\ &= \left\{ v \in \mathbb{R}^n : \sum_{i=1}^n v_i = 0 \right\} \\ &= \left\{ v \in \mathbb{R}^n : \bar{v} = 0 \right\} \end{aligned}$$

$$\tilde{X}_2 = M_\ell X_2$$

- Mean of the regressor:

$$\begin{aligned} (\ell^\top \ell)^{-1} \ell^\top X_2 &= \frac{\ell^\top X_2}{n} \\ &= \frac{1}{n} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}^\top \begin{bmatrix} X_{1,2} \\ \vdots \\ X_{n,2} \end{bmatrix} \\ &= \frac{1}{n} \sum_{i=1}^n 1 \cdot X_{i,2} \\ &= \frac{1}{n} \sum_{i=1}^n X_{i,2} \\ &= \bar{X}_2 \end{aligned}$$

- The regressor:

$$\begin{aligned} \tilde{X}_2 &= M_\ell X_2 \\ &= X_2 - \ell \frac{\ell^\top X_2}{n} \\ &= X_2 - \bar{X}_2 \ell \\ &= \begin{pmatrix} X_{1,2} - \bar{X}_2 \\ \vdots \\ X_{n,2} - \bar{X}_2 \end{pmatrix} \end{aligned}$$

- The dependent variable: $M_\ell Y = (Y_1 - \bar{Y}, \dots, Y_n - \bar{Y})^\top$.

Slope estimator and deviations from the average

- We will use: $M_\ell^\top = M_\ell$, $M_\ell M_\ell = M_\ell$, and $X_2^\top M_\ell X_2$: 1×1 .

- We have:

$$\begin{aligned}
\hat{\beta}_2 &= (X_2^\top M_\ell X_2)^{-1} X_2^\top M_\ell Y \\
&= \frac{X_2^\top M_\ell Y}{X_2^\top M_\ell X_2} \\
&= \frac{(M_\ell X_2)^\top Y}{(M_\ell X_2)^\top (M_\ell X_2)} \\
&= \frac{\tilde{X}_2^\top Y}{\tilde{X}_2^\top \tilde{X}_2} \\
&= \frac{(X_2 - \bar{X}_2 \ell)^\top Y}{(X_2 - \bar{X}_2 \ell)^\top (X_2 - \bar{X}_2 \ell)} \\
&= \frac{\sum_{i=1}^n (X_{i,2} - \bar{X}_2) Y_i}{\sum_{i=1}^n (X_{i,2} - \bar{X}_2)^2} \\
&= \boxed{\frac{\sum_{i=1}^n (X_{i,2} - \bar{X}_2)(Y_i - \bar{Y})}{\sum_{i=1}^n (X_{i,2} - \bar{X}_2)^2}} \quad (X_2^\top M_\ell Y = X_2^\top M_\ell^\top M_\ell Y)
\end{aligned}$$

Several regressors and an intercept

- Intercept β_1 and slope coefficients β_2 :

$$X = \begin{bmatrix} \ell & X_2 \\ n \times k & n \times 1 & n \times (k-1) \end{bmatrix}, \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}, \quad Y = \beta_1 \ell + X_2 \beta_2 + U$$

- Each regressor minus its mean:

$$\begin{aligned}
\tilde{X}_2 &= M_\ell X_2 \\
&= M_\ell [\underline{X}_2 \cdots \underline{X}_k] \\
&= [M_\ell \underline{X}_2 \cdots M_\ell \underline{X}_k] \\
&= [\tilde{\underline{X}}_2 \cdots \tilde{\underline{X}}_k]
\end{aligned}$$

- Slope coefficients:

$$\begin{aligned}
\hat{\beta}_2 &= (X_2^\top M_\ell X_2)^{-1} X_2^\top M_\ell Y \\
&= (\tilde{X}_2^\top \tilde{X}_2)^{-1} \tilde{X}_2^\top Y
\end{aligned}$$

- Variation of X_2 in the sample, with $X_{i,2}$ now $(k-1) \times 1$ and $\bar{X}_2 = \frac{1}{n} \sum_{i=1}^n X_{i,2}$:

$$\begin{aligned}
\tilde{X}_2^\top \tilde{X}_2 &= \sum_{i=1}^n \tilde{X}_{i,2} \tilde{X}_{i,2}^\top \\
&= \sum_{i=1}^n (X_{i,2} - \bar{X}_2)(X_{i,2} - \bar{X}_2)^\top
\end{aligned}$$

Summary

Projections

- $P_X = X(X^\top X)^{-1} X^\top$ and $M_X = I_n - P_X$: symmetric, idempotent, p.s.d.
- $P_X X = X$, $M_X X = 0$, $P_X M_X = 0$.
- $\text{tr}(P_X) = k$ and $\text{rank}(M_X) = \text{tr}(M_X) = n - k$.
- $Y = \hat{Y} + \hat{U}$ with $\hat{Y} = P_X Y \in S(X)$, $\hat{U} = M_X Y \in S^\perp(X)$, $\hat{Y}^\top \hat{U} = 0$.

s^2 and partitioned regression

- $\hat{U} = M_X U$, $\hat{U}^\top \hat{U} = U^\top M_X U$, $E[\hat{U}^\top \hat{U} \mid X] = \sigma^2 \text{tr}(M_X) = \sigma^2(n - k)$.
- $E[\hat{\sigma}^2 \mid X] = \frac{n-k}{n} \sigma^2$; $s^2 = \hat{U}^\top \hat{U} / (n - k)$ with $E[s^2 \mid X] = \sigma^2$; $\widehat{\text{Var}}(\hat{\beta} \mid X) = s^2(X^\top X)^{-1}$.
- Frisch-Waugh-Lovell: $\hat{\beta}_1 = (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 Y = (\tilde{X}_1^\top \tilde{X}_1)^{-1} \tilde{X}_1^\top \tilde{Y}$, with $\tilde{X}_1 = M_2 X_1$ and $\tilde{Y} = M_2 Y$.
- Intercept, one regressor: $\hat{\beta}_2 = \sum_i (X_{i,2} - \bar{X}_2) Y_i / \sum_i (X_{i,2} - \bar{X}_2)^2$.